

Yoonali

Philadelphia, PA | (215) 869-5600 | yoonali@engineering.upenn.edu | Portfolio

EDUCATION

University of Pennsylvania, Penn Engineering

Master of Applied Science in Computer Science (MAS-CS)

New York University

Bachelor of Economics & Statistics

Philadelphia, PA

Expected May 2028

New York, NY

May 2026

TECHNICAL SKILLS

Programming: Python, Java, C, SQL, R, MATLAB

Frameworks & Libraries: FastAPI, React, PyTorch, scikit-learn, XGBoost, NumPy, Pandas

Machine Learning: Random Forest, classification, fine-tuning, feature engineering, SHAP/LIME, model evaluation, latency/accuracy benchmarking

LLM / AI Systems: RAG, vector retrieval, streaming chat, prompt engineering, LLM evaluation, LoRA/LoRA+/QLoRA, retrieval-aware generation/abstention, LiveKit/TTS

Developer & Data Tools: Git, Docker, GitHub Actions (CI/CD); data pipelines, stateful conversational systems, reproducible testing, annotation/QC

EXPERIENCE

Lumi AI

Founder & Technical Product Lead

Philadelphia, PA

Jun 2026 - Present

- Adapted and extended a team-developed React/FastAPI AI tutoring platform into a teacher-grounded learning product, connecting file ingestion, vector retrieval, streamed chat, bot configuration, and real-time voice to child-facing practice and parent-facing evidence.
- Implemented retrieval-aware response boundaries and abstention behavior to reduce unsupported answers; structured retrieval, generation, safety, and evaluation as modular components for reproducible iteration across configurations.
- Built Test & Compare workflows across retrieval, prompt, and response variants, evaluating groundedness, uncertainty calibration, age-appropriate behavior, evidence of understanding, and response quality rather than relying on single prompt-level checks.
- Translated educator and preschool feedback into technical requirements for pilot workflows, product iteration, and reliability constraints, connecting software decisions to user needs and commercialization hypotheses.

ByteDance (TikTok)

AI Product Manager Intern - Search Ads

Beijing, China

Jun 2026 - Aug 2026

- Designed and evaluated query-similarity routing and dynamic-threshold strategies across recall, ad load, and auction quality, contributing more than \$145K in incremental revenue while preserving organic-search relevance.
- Prototyped an LLM-powered Q&A advertising experience and defined evaluation logic linking answer-seeking intent, query relevance, generated-content quality, conversion, and overall search quality.

ThunderSoft

Algorithm Intern

Beijing, China

Jun 2025 - Jul 2025

- Built a 100K+ sample annotation/QC pipeline for a Xiaomi mobile-AI project, defining taxonomy, quality-control rules, annotator guidance, and reproducible checks to improve dataset consistency.
- Fine-tuned PyTorch models and ran reproducible latency/accuracy benchmarks, improving accuracy by approximately 8%, reducing inference latency by approximately 5%, and increasing evaluation efficiency by approximately 20%.

PROJECTS & PUBLICATION

FashionVote

Founder & Research Lead

Aug 2025 - Dec 2025

- Built and deployed an AI-driven fashion analytics platform using DeepFashion imagery and a crowdsourced pairwise-voting interface; curated and deduplicated image pairs and trained Random Forest models achieving 90%+ accuracy in trend classification.
- Analyzed preference data from 10K+ users, diagnosed sampling bias and label sparsity, and narrowed model interpretation toward defensible user-level recommendations; translated findings into retail product and marketing discussions.

Unified Parameter-Efficient Adaptation Framework for Quantized LLMs

Sole Author, IJNSEI

2025

- Developed a unified parameter-efficient fine-tuning framework integrating LoRA, LoRA+, and QLoRA for 4-bit quantized decoder-only LLMs, enabling billion-parameter adaptation with constrained compute.
- Built reproducible PyTorch benchmarks across trainable parameters, memory footprint, and task performance; demonstrated comparable adaptation quality with approximately 1% trainable parameters.

RESEARCH EXPERIENCE

The Wharton School

Research Assistant, Prof. Jonah Berger

Philadelphia, PA

Jun 2026 - Aug 2026

- Operationalized curiosity and information-gap theory into a reproducible behavioral-data annotation framework with a calibrated 7-point schema, explicit decision rules, edge-case taxonomy, and adjudication logic.
- Reviewed ambiguous and borderline cases and documented resolution criteria to improve consistency and reproducibility of downstream analysis.

NYU School of Global Public Health

Biostatistics Research Assistant

New York, NY

Sep 2025 - May 2026

- Designed child/family feedback surveys and a small-N ($N < 200$) tracking framework for pediatric psychological evaluations, translating clinical questions into structured binary, ordinal, categorical, and qualitative variables.
- Structured data for interpretable comparisons of length of stay by diagnosis and evaluation type while explicitly accounting for small-sample limits on inference.

NYU School of Global Public Health

Research Assistant, Public Health Backlash Lab

New York, NY

Mar 2025 - May 2026

- Collaborated with a cross-institutional NYU/Ohio State/University of Minnesota team; conducted Dedoose coding and analytic memoing and synthesized mixed-methods evidence on communication, policy response, community reaction, and public trust.

ENTREPRENEURSHIP & LEADERSHIP

HaoXue AI Learning Center

Founder

May 2023 - Present

- Founded and scaled an offline AI-powered learning and enrichment center serving 500+ students; led program design, student growth, parent engagement, and day-to-day operations.

UNICEF at NYU

Founder & President

New York, NY

Mar 2025 - May 2026

- Founded NYU's first UNICEF student chapter, led a 20-person team, and delivered programming including a 150+ attendee flagship event; managed outreach, partnerships, speaker coordination, and execution.